

VENKATESWARARAO PERAM

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C++ Developer | RTOS | Unit Testing | 3 Years Experience

PROFESSIONAL SUMMARY

- C++ Developer with around 3 years of experience in application-level C++ development for real-time systems.
- Hands-on experience working on **RTOS-based platforms (QNX)**, focusing on runtime behavior, stability, and debugging.
- Strong experience in **unit testing as part of development**, dynamic testing, requirement-based and scenario-based testing.
- Worked extensively on **Qt**-based HMI applications (IPC, RCC), validating UI logic, signal-driven behavior, and real-time updates.
- Actively involved in **code reviews (PRs)** and ensured compliance with **AUTOSAR C++14** and **MISRA** guidelines.
- Proven ability to manage competing priorities, meet sprint goals, and deliver high-quality software in Agile teams.

SKILLS & EXPERTISE

Programming: C++, OOP

HMI Frameworks: Qt

Real-Time Systems: windows /QNX RTOS

Testing: Unit Testing, Dynamic Testing, Requirement-Based Testing, Scenario-Based Testing

Testing Tools: Vector CAST, Para Soft

Code Quality: AUTOSAR C++14, MISRA C++, Static Code Analysis, PR Reviews

CI/CD & Build: Git, Bitbucket/GitHub, Jenkins, CMake

Process: Agile/Scrum, Defect Tracking, Root Cause Analysis

PROFESSIONAL EXPERIENCE

Associate Software Engineer | People Tech Group (Client: General Motors)

April 2023 – Present

- Developed and maintained C++ application modules for real-time automotive systems running on QNX RTOS.
- Implemented application logic using modern C++ with focus on performance and stability.
- Performed **unit testing as an integral part of development** using, Vector CAST, and ParaSoft Unit.
- Executed **dynamic testing** on RTOS targets to validate runtime behavior, task interactions, and timing constraints.
- Investigated defects through debugging and root-cause analysis; implemented fixes and validated solutions.
- Reviewed pull requests and ensured adherence to **AUTOSAR** and **MISRA** coding standards.
- Managed multiple tasks in parallel, met sprint goals, and supported action plans to achieve project objectives.
- Collaborated with cross-functional teams in Agile/Scrum environments.

PROJECT EXPERIENCE

RTOS-Based HMI Application – Instrument Panel Cluster (IPC)

- Developed and tested C++ modules for speed, fuel, warnings, and tell-tales on QNX RTOS.
- Validated real-time behavior, data updates, and display timing through dynamic testing.
- Debugged and resolved issues related to UI refresh and data consistency.

RTOS-Based HMI Application – Front & Rear Command Console (RCC)

- Developed backend C++ logic for climate and comfort features integrated with HMI layers.
- Performed scenario-based and regression testing for user interactions and system responses.
- Fixed defects and validated stability across multiple software builds.

EDUCATION

Bachelor of Technology in Electronics and Communication Engineering (ECE)

SSIET Group of Institutions – 2020